ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

PROJECT MANAGEMENT

PRACTICE EXAM · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Project Management practice exam.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer. Use consistent accuracy. Answers must relate to the case study where asked; accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 D ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 D ✓ 1.1.8 B ✓ 1.1.9 D ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 E ✓ 1.3.2 D ✓ 1.3.3 B ✓ 1.3.4 G ✓ 1.3.5 H ✓ 1.3.6 I ✓ 1.3.7 F ✓ 1.3.8 J ✓ 1.3.9 C ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FOUR: no long queues – students register from anywhere ✓; faster processing / fewer staff hours ✓; fewer capturing errors ✓; better reports and planning (student numbers per course) ✓; less paper and storage; 24-hour access; improved college image (4)

2.2.1 $\text{Option A: } 400\,000 \div 160\,000 \checkmark = 2{,}5 \text{ years} \checkmark; \text{ Option B: } 300\,000 \div 90\,000 \checkmark = 3{,}33 \text{ years} \checkmark$ (4)

2.2.2 $\text{Savings over } 5 \text{ years} = 5 \times 160\,000 = \text{R}800\,000 \checkmark; \text{ ROI} = (800\,000 - 400\,000) \div 400\,000 \times 100 \checkmark = 100\% \checkmark$ (3)

2.2.3 Option A ✓ – shorter payback period / higher savings ✓; non-financial reason, e.g. it serves more students and reduces queues more / better fits the college's strategy ✓ (accept B with valid reasons, e.g. lower cost if the budget is limited) (3)

2.3 Any TWO: fit with the organisation's strategy / goals ✓; legal or compliance requirements ✓; risk level; social or environmental impact; available skills and resources; urgency (2)

[16]

QUESTION 3

3.1.1 $\text{EV} = 40\% \times \text{R}400\,000 \checkmark = \text{R}160\,000 \checkmark$ (2)

3.1.2 $\text{SV} = \text{EV} - \text{PV} = 160\,000 - 200\,000 \checkmark = -\text{R}40\,000 \checkmark; \text{CV} = \text{EV} - \text{AC} = 160\,000 - 180\,000 \checkmark = -\text{R}20\,000 \checkmark$ (4)

3.1.3 $\text{SPI} = \text{EV} \div \text{PV} = 160\,000 \div 200\,000 \checkmark = 0{,}80 \checkmark$; $\text{CPI} = \text{EV} \div \text{AC} = 160\,000 \div 180\,000 \checkmark = 0{,}89 \checkmark$ (4)

3.1.4 Behind schedule (SV negative, SPI < 1) ✓ and over budget (CV negative, CPI < 1) ✓ (2)

3.1.5 $\text{EAC} = 400\,000 \div 0{,}89 \checkmark = \text{R}449\,438{,}20 (\approx \text{R}450\,000 \text{ using CPI } 0{,}8889) \checkmark$ (2)

3.2 Any TWO: add resources (crashing) ✓; do activities in parallel (fast-tracking) ✓; overtime; reduce scope with the sponsor's approval; remove obstacles / improve productivity (2)

[16]

QUESTION 4

4.1 3 marks per risk (risk ✓, sensible rating ✓, response ✓), any FOUR: system crashes during peak registration (medium probability / high impact) – load testing, extra server capacity; developer delivers late (medium/high) – penalty clauses, milestones; students without Internet access (high/medium) – computer kiosks at campus; staff resistance to change (medium/medium) – training and communication; data breach of personal information (low/high) – encryption, security testing, POPIA compliance; budget overrun (medium/high) – contingency, cost control (12)

4.2 Quality assurance – planned, process-focused activities that prevent defects ✓, e.g. agreed coding standards, regular reviews of the developer's work ✓; quality control – inspecting / testing deliverables to find defects ✓, e.g. testing the registration form with real student data ✓ (4)

4.3 Any FOUR: price ✓; experience and track record / references ✓; technical solution meets requirements ✓; delivery time ✓; support and maintenance offered; B-BBEE status; financial stability; security and data protection (4)

[20]

QUESTION 5

5.1 Forming – members meet, polite, unsure of roles ✓✓; storming – conflict over ideas and roles ✓✓; norming – agreement on rules and ways of working, cooperation grows ✓✓; performing – the team works effectively towards the goals ✓✓ (8)

5.2 2 marks per stakeholder (information ✓, method and frequency ✓), e.g. principal / steering committee – progress, budget, risks; monthly report and meeting; developers – requirements, changes, test results; weekly meetings / e-mail; students – registration dates and instructions; SMS, website, social media before and during January; IT department – server requirements, schedule; weekly meetings (6)

5.3 It evaluates the project after it is finished to see whether it achieved its objectives and to learn lessons for future projects ✓✓. Any TWO questions ✓✓: Was it on time and within budget? Did the system reduce queues / meet the users' needs? What went well and what went wrong? Are users satisfied? (4)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE